

Personal reminiscences of Bruce Francis

from Tryphon Georgiou

First came to know Bruce
at the 25th CDC in Athens, Greece

I visited him Toronto in early 1987
and initially, the plan by Bruce and me,
was for us to think through on
how to work out
robust control for sampled data systems

but it was too early to deal with norms,
and we scaled back,
to focus on gain margin!

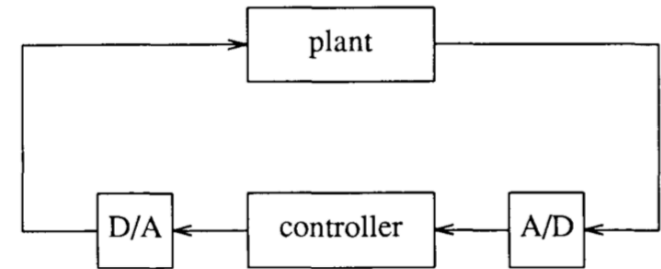
This was the start of a collaboration
and a deep friendship.



Toronto 1987 - in Bruce's office

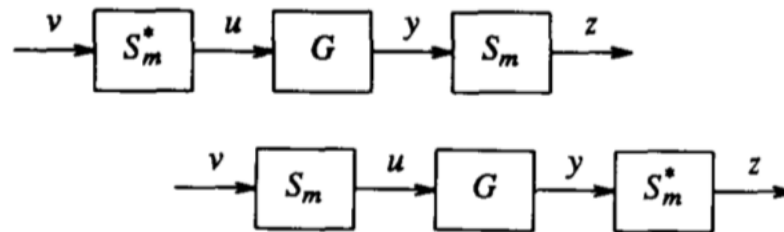
Stability Theory for Linear Time-Invariant Plants with Periodic Digital Controllers

BRUCE A. FRANCIS, AND TRYPHON T. GEORGIOU



Bruce's perfectionist style
led us systematically to lay out stability theory results first
for subsampled period as well as continuous-sampled-data

honing in an operator framework
and lifting techniques, already in use in periodic control,
and functional analysis

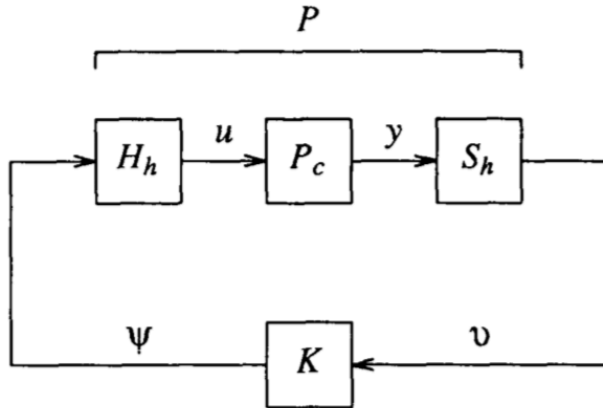


then exploited the ability of periodic control
to alter the transmission-zero structure of systems

building on
Tannenbaum's gm problem and
Khargonekar, Poola, Tannenbaum

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Theorem 6: For every interval (a, b) , $0 < a < b < \infty$, there exist a sampling period h and a periodic causal controller K (that is implemented in the form $H_h S_m^* L S_m S_h$) such that $(kP_c; K)$ is stable for all k in (a, b) .

...

theory of hybrid systems, constructive state-space techniques for assigning zeros using periodic digital control, and its application to the gain margin optimization of continuous-time systems.

However, questions regarding the performance of digital feedback controllers, as well as the possible limitation of periodic feedback control in general, remain important subjects for further research. For instance, it is not known how to compare quantitatively the improvement in the sensitivity with respect to a certain parameter against a potential deterioration in intersample behavior, disturbance attenuation, or even possible increase in the sensitivity to other parameters.



From Bruce:

Friday, 13 April, 2012

My research career

A biographical sketch for my own amusement.

Ansari

I studied mechanical engineering at U of T from 1965 to 1969. In my final undergraduate year one of my courses was Control Systems. The professor was a man from India named J. S. Ansari.

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The text for Ansari's course was *Control Systems*. Over the years I've come to realize that, like all the other texts of the time, is a dreadful book. The models are trivial and not related to the real object; control specifications are not motivated -- you have to get 60 degrees phase margin but you don't know why; performance limitations are non-existent; and the important results of Bode, the theoretical underpinning of classical control, are neglected. Nevertheless, because it was simple and largely methodical I liked the subject and I did an M. Eng. degree under the supervision of Professor Ansari; my project involved a heat-exchange system with a pneumatic PID controller. I learned nothing from the project.

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Found in Sent - umn Mailbox



Tryphon Georgiou

March 13, 2013 at 7:36 PM



Pope Francis --- congratulations Bruce!! :-)

[Details](#)

To: Francis bruce, Cc: Allen Tannenbaum

<http://www.cbc.ca/news/world/story/2013/03/13/vatican-pope-francis-profile.html>

Francis bruce

March 14, 2013 at 2:47 AM



Re: Pope Francis --- congratulations Bruce!! :-)

To: Tryphon Georgiou

Bless you, my son.

Early 1990's - two plates!!



MTNS 1989



Bruce Francis
a scholar and a gentleman
and a dear friend

Netherlands 1990's

